

### 3D Geometry

**Problem 1.** Given a convex polyhedron, the angular defect at a vertex is defined as  $360^\circ$  minus the sum of angles at this vertex. Prove that the total angular defect of a convex polyhedron is  $720^\circ$ .

**Problem 2.** Let  $ABCD$  be a tetrahedron such that  $\angle BAC = \angle ACD$  and  $\angle ABD = \angle BDC$ . Prove that  $AB = CD$ .

**Problem 3.** The incircles of the faces  $ABC$  and  $ABD$  of a tetrahedron  $ABCD$  are tangent to the edge  $AB$  in the same point. Prove that the points of tangency of these incircles to the edges  $AC, BC, AD, BD$  are concyclic.

**Problem 4.** Let  $ABCD$  be a tetrahedron. Suppose the medians from  $A$  of  $\triangle ABC$ ,  $\triangle ABD$  and  $\triangle ACD$  are pairwise perpendicular. Prove that  $AB = AC = AD$ .

**Problem 5.** Let  $P_1P_2P_3P_4$  be a tetrahedron in  $\mathbb{R}^3$  and let  $O$  be a point equidistant from each of its vertices. Suppose there exists a point  $H$  such that for each  $i$ , the line  $P_iH$  is perpendicular to the plane through the other three vertices. Line  $P_1H$  intersects the plane through  $P_2, P_3, P_4$  at  $A$ , and contains a point  $B \neq P_1$  such that  $OP_1 = OB$ . Show that  $HB = 3HA$ .

**Problem 6.** The sphere inscribed in tetrahedron  $ABCD$  touches the sides  $ABD$  and  $DBC$  at points  $K$  and  $M$ , respectively. Prove that  $\angle AKB = \angle DMC$ .

**Problem 7.** Let  $ABCD$  be an orthocentric tetrahedron with orthocentre  $H$ . Prove that  $HBCD$  is orthocentric, and  $A$  is the orthocentre of  $HBCD$ .

**Problem 8.** Let  $ABCD$  be an orthocentric tetrahedron. Let  $\ell_A$  be the line passing through the nine-point centre of  $\triangle BCD$  and perpendicular to the plane  $BCD$ . Similarly, define  $\ell_B, \ell_C$  and  $\ell_D$ . Prove that  $\ell_A, \ell_B, \ell_C$  and  $\ell_D$  are concurrent.

**Problem 9.** Perpendiculars  $AB_1, AC_1, AD_1$  were dropped in the tetrahedron  $ABCD$  from the vertex  $A$  on the plane dividing the dihedral angles at the edges  $CD, BD, BC$  in half. Prove that the plane  $B_1C_1D_1$  is parallel to the plane  $BCD$ .

**Problem 10.** Let  $ABCD$  be an orthocentric tetrahedron. Let  $H_B, H_C, H_D$  be the feet of altitudes from  $B, C, D$  respectively. Prove that  $B, C, D, H_B, H_C, H_D$  are cospherical.

**Problem 11.** Point  $M$  is marked on a sphere. Consider all triples of points  $A, B, C$  on the sphere other than  $M$ , such that the segments  $MA, MB, MC$  are pairwise perpendicular, and for each such triple we consider the plane passing through  $A, B, C$ . Prove that all such planes pass through a common point.

**Problem 12.** Let  $R$  and  $r$  be the circumradius and inradius of a tetrahedron  $ABCD$ . Prove that  $R \geq 3r$ .

**Problem 13.** The insphere of a tetrahedron  $ABCD$  touches its faces at points  $A', B', C', D'$  respectively. If the segments  $AA'$  and  $BB'$  intersect, prove that the segments  $CC'$  and  $DD'$  also intersect.

**Problem 14.** Let  $I$  be the incentre of a tetrahedron  $ABCD$ . The insphere of  $ABCD$  touches the faces  $ABC, ABD, ACD$  at  $T_D, T_C, T_B$  respectively. The plane  $T_B T_C T_D$  cuts the edges  $AB, AC, AD$  at  $X_B, X_C, X_D$  respectively. Prove that the incentre of  $AX_B X_C X_D$  is the intersection of the segment  $AI$  and the insphere of  $ABCD$ .

**Problem 15.** Let  $I$  be the incentre of a tetrahedron  $ABCD$ . The insphere touches the plane  $BCD$  at  $T$ . Let  $I'$  be the antipode of  $I$  on the circumsphere of  $IBCD$ . The line  $I'T$  meets the insphere of  $ABCD$  again at  $P$ . Prove that the circumsphere of  $PBCD$  is tangent to the insphere of  $ABCD$ .

**Problem 16.** The insphere and the exsphere opposite to the vertex  $D$  of a tetrahedron  $ABCD$  touch the face  $ABC$  in the points  $X$  and  $Y$ , respectively. Show that  $\angle XAB = \angle CAY$ .

**Problem 17.** Let  $ABCD$  be a tetrahedron. The points  $P_{AB}, P_{AC}, P_{AD}, P_{BC}, P_{BD}$  and  $P_{CD}$  are chosen on the edges  $AB, AC, AD, BC, BD, CD$  respectively. Prove that the circumspheres of  $AP_{AB}P_{AC}P_{AD}, BP_{AB}P_{BC}P_{BD}, CP_{AC}P_{BC}P_{CD}, DP_{AD}P_{BD}P_{CD}$  has a common point.

**Problem 18.** A regular spatial pentagon consists of five points  $P_1, P_2, P_3, P_4, P_5$  in  $\mathbb{R}^3$  such that  $|P_i P_{i+1}| = |P_j P_{j+1}|$  and  $\angle P_{i-1} P_i P_{i+1} = \angle P_{j-1} P_j P_{j+1}$  for all  $1 \leq i, j \leq 5$ , where  $P_0 = P_5$  and  $P_6 = P_1$ . A regular spatial pentagon is planar if there is a plane passing through all five points  $P_1, P_2, P_3, P_4, P_5$ .

Show that every regular spatial pentagon is planar.

## Combinatorial Geometry

**Problem 19.** There are 2020 different planes in the space. Prove that there exists a sphere intersecting exactly 100 of these planes.

**Problem 20.** There are 2020 distinct points in  $\mathbb{R}^3$ . Prove that there are two non-intersecting spheres each containing exactly 1010 of these points.

**Problem 21.** Given any five points on a sphere, show that some four of them must lie on a closed hemisphere.

**Problem 22.** Consider a convex polyhedron with at least five faces such that exactly three edges emerge from each of its vertices. Two players play the following game: Each player, in turn, signs his or her name on a previously unsigned face. The winner is the player who first succeeds in signing three faces that share a common vertex.

Show that the player who signs first will always win by playing as well as possible.

**Problem 23.** A sphere is completely covered by a finite number of (possibly overlapping) hemispheres. Prove that we can choose at most 4 of these hemispheres which together cover the sphere completely.

**Problem 24.** There is a set  $S$  of  $n \geq 4$  points in  $\mathbb{R}^3$  such that no 4 points are coplanar. Suppose we can colour all the points in  $S$  black and white such that for every sphere passing through at least 4 points in  $S$ , exactly half of these common points are coloured white. Prove that all points in  $S$  are cospherical.